STAT 1210: Assignment 3 Solutions

Part III: Foundations of Inference

Fall 2026 Semester

Suggested Grading

Question	Marks by part	Chapter focus	Total
Question 1	$4+4+4+4+4$	Study design	20
Question 2	$4+4+4+4+4$	Sampling distributions	20
Question 3	$4+4+4+4+4$	Confidence intervals	20
Question 4	$4+4+4+4+4$	Hypothesis testing	20
Question 5	$4+4+4+4+4$	Inference in practice	20
		Total	100

Question 1

1. The individuals are the 600 participating adult dogs. The intended population is all adult dogs in Prince Edward Island. The sample consists of the 600 dogs whose owners participated at the five volunteer clinics during the two-week period.

   The two variables are daily activity group and obesity status. Both are categorical because the recorded values are categories rather than numerical measurements.

2. For dogs with at least 60 minutes of activity,

$$\frac{18}{150} = 0.120$$

   For dogs with less than 60 minutes of activity,

$$\frac{126}{450} = 0.280$$

   The obesity proportion is 0.160, or 16 percentage points, higher in the lower-activity group. The sample therefore shows an association between lower activity and obesity.

3. This is an observational study because the researchers recorded existing characteristics without assigning a treatment. Random sampling was not used because clinics volunteered, the dogs were limited to appointment attendees and owners chose whether to participate. Random assignment was not used because dogs were not assigned to activity levels.

4. Two plausible confounders are age and diet. Older dogs may be less active and may also have a greater obesity risk. Dogs receiving more food or high-calorie treats may have fewer exercise opportunities, while greater calorie intake directly increases obesity risk. Other reasonable answers include breed, illness, medication, neuter status and owner lifestyle.

5. Volunteer clinics may differ from nonparticipating clinics, and dogs that rarely receive veterinary care are undercovered. Owner participation can also create voluntary-response or nonresponse bias.

   The study cannot establish causation because there was no random assignment and confounding remains possible. It also cannot be confidently generalized to all adult dogs in Prince Edward Island because neither clinics nor dogs were selected through probability sampling.

Question 2

1. The population parameter is $p = 0.42$, the probability that a tagged juvenile Atlantic salmon survives the specified route under similar conditions. The statistic is

$$\hat{p} = \frac{\text{number of sampled salmon that survive}}{400}$$

One salmon's outcome is categorical because it is survival or no survival. The sample proportion is quantitative because it is a numerical summary between 0 and 1.

2. The sampling-distribution mean is

$$\mu_{\hat{p}} = p = 0.42$$

The standard deviation is

$$\sigma_{\hat{p}} = \sqrt{\frac{0.42(0.58)}{400}} \approx 0.02468$$

Across repeated random samples of 400 salmon, the sample proportions are centred at 0.42 and typically vary from 0.42 by about 0.0247.

3. The expected counts are

$$np = 400(0.42) = 168 \qquad n(1-p) = 400(0.58) = 232$$

Both are at least 10. Together with the stated independence condition, this supports the Normal approximation.

4. The standardized value is

$$z = \frac{0.38 - 0.42}{0.02468} \approx -1.621$$

Therefore,

$$P(\hat{p} \leq 0.38) \approx P(Z \leq -1.621) \approx 0.0525$$

5. The standardized boundaries are

$$z_L = \frac{0.40 - 0.42}{0.02468} \approx -0.810 \qquad z_U = \frac{0.46 - 0.42}{0.02468} \approx 1.621$$

Thus,

$$P(0.40 \leq \hat{p} \leq 0.46) \approx \Phi(1.621) - \Phi(-0.810) \approx 0.7386$$

Because $\sigma_{\hat{p}}$ is proportional to $1/\sqrt{n}$, halving it requires four times the sample size:

$$n_{\text{new}} = 4(400) = 1600$$

Question 3

1. The parameter μ is the population mean recovery time for all adult dogs receiving the routine procedure in the hospital network during the period of interest. Its point estimate is $\bar{x} = 47.8$ minutes.

2. The sample is described as a simple random sample, and recovery times from different dogs must be sufficiently independent. Since the population exceeds 1000 dogs,

$$64 < 0.10(1000) = 100$$

   so the 10% condition is satisfied. With $n = 64$, the Central Limit Theorem generally supports an approximately Normal sampling distribution unless the population is extremely skewed or has severe outliers. Those features cannot be checked without the observations or a graph.

3. The standard error and margin of error are

$$SE = \frac{12}{\sqrt{64}} = 1.5$$

$$m = 1.960(1.5) = 2.94$$

   The 95% confidence interval is

$$\boxed{47.8 \pm 2.94 = (44.86,\ 50.74) \text{ minutes}}$$

4. We are 95% confident that the population mean recovery time for adult dogs receiving this procedure in the hospital network is between approximately 44.86 and 50.74 minutes. The interval concerns the population mean, not the recovery times of 95% of individual dogs.

5. For 99% confidence,

$$m_{99} = 2.576(1.5) = 3.864$$

$$\boxed{47.8 \pm 3.864 = (43.936,\ 51.664) \text{ minutes}}$$

   The 95% interval has width 5.88 minutes, whereas the 99% interval has width 7.728 minutes. Greater confidence requires a larger critical value and therefore a wider interval.

Question 4

1. The parameter μ is the population mean resting respiratory rate for adult harbour seals recovering under the revised protocol. The hypotheses are

$$H_0 : \mu = 18 \qquad H_a : \mu \neq 18$$

The alternative is two-sided because rates below or above 18 may both be clinically important.

2. The observations sum to 503, so

$$\bar{x} = \frac{503}{25} = 20.12$$

The standard error is

$$SE = \frac{4}{\sqrt{25}} = 0.8$$

The test statistic is

$$z = \frac{20.12 - 18}{0.8} = 2.650$$

The sample mean is 2.65 standard errors above the null value.

3. From the Standard Normal table,

$$P(Z \geq 2.65) \approx 0.0040$$

The two-sided P-value is

$$2P(Z \geq 2.65) \approx 2(0.0040) = 0.0080$$

Since $0.0080 < 0.05$, reject H_0. The data provide statistically significant evidence that the population mean respiratory rate under the revised protocol differs from 18 breaths per minute.

4. A Type I error means concluding that the revised-protocol mean differs from 18 when the true mean is 18. A Type II error means failing to detect a genuine difference. A Type II error may be especially serious if an unsafe protocol is retained, although a Type I error could also matter if a safe protocol is abandoned. Either choice is acceptable with a clear animal-safety justification.

5. The 95% interval is

$$20.12 \pm 1.960(0.8)$$

$$\boxed{(18.552,\ 21.688) \text{ breaths per minute}}$$

The null value 18 is outside the interval, agreeing with rejection by the corresponding two-sided test. Statistical significance does not establish clinical importance; the team should also consider the estimated effect size, a clinically meaningful threshold, adverse events and other recovery outcomes.

Question 5

1. The parameter μ is the population mean portable-meter reading minus reference-laboratory reading for canine blood samples in the population represented by the study. The standard error and test statistic are

$$SE = \frac{2.4}{\sqrt{400}} = 0.12 \qquad z = \frac{0.30}{0.12} = 2.500$$

   The two-sided P-value is

$$2P(Z \geq 2.50) = 2(0.0062) \approx 0.0124$$

   Since $0.0124 < 0.05$, reject H_0. The sample provides statistically significant evidence of a nonzero population mean difference.

2. The 95% margin of error is

$$m_{95} = 1.960(0.12) = 0.2352$$

   The interval is

$$\boxed{0.30 \pm 0.2352 = (0.0648,\ 0.5352)\ \text{mg/dL}}$$

   The interval excludes 0, agreeing with rejection by the two-sided test at $\alpha = 0.05$.

3. The finding is statistically significant, but the point estimate is only 0.30 mg/dL and the entire 95% interval lies between -1.0 and 1.0 mg/dL. According to the laboratory's threshold, the plausible mean biases are too small to be practically important.

   A possible source of systematic error is miscalibration of the reference instrument, sample degradation during storage or undercoverage from using only archived network samples. The margin of error measures random sampling variability, not systematic measurement or selection bias.

4. For 99% confidence,

$$m_{99} = 2.576(0.12) = 0.30912$$

$$\boxed{0.30 \pm 0.30912 = (-0.00912,\ 0.60912)\ \text{mg/dL}}$$

   The interval is wider and includes 0. The corresponding test at $\alpha = 0.01$ fails to reject H_0, consistent with $0.0124 > 0.01$.

5. The required sample size is

$$n = \left(\frac{(1.645 + 1.282)(2.4)}{0.6}\right)^2 \approx 137.077$$

   Therefore,

$$\boxed{n = 138}$$

   The value is rounded up so that the planned power is at least 90%.